

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA1603 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A MCQ TEST

<i>Criteria</i>	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	<i>Total(10)</i>
<i>Weightage</i>	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I M A Surya Kiran (192411236) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M A Surya Kiran

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand



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Think Before You Decide: Inside Decision Support Systems

CSA1603 –Data Warehousing And Data Mining

Course Faculty :

Beulah David D

M Aslesha Surya Kiran (192411236)

INTRODUCTION

- Decision-making is an essential part of everyday life and business operations.
- Organizations generate large volumes of data that can be difficult to analyze manually.
- A Decision Support System (DSS) is a computer-based system that helps users make informed and effective decisions.
- DSS combines data, analytical models, and user-friendly software to provide valuable insights.
- DSS is widely used in healthcare, banking, agriculture, education, business, and transportation.

WORKING PROCESS:

- Data Collection
- Data Analysis
- Analytical Models
- Generate Insights
- Decision Making

Example

A bank uses DSS to approve loans by analyzing:

- Customer income
- Credit score
- Loan history
- Risk level

Outcome: The system recommends whether to approve or reject the loan.

Applications of DSS

Industry	Application
 Healthcare	Disease diagnosis & treatment planning
 Banking	Loan approval & fraud detection
 Agriculture	Crop prediction & irrigation planning
 Transportation	Route optimization
 E-Commerce	Product recommendations
 Education	Student performance analysis

Benefits&Challenges

Benefits:

- Faster decision-making
- Improved accuracy
- Reduces human errors
- Better planning
- Increases productivity.

Challenges:

- Poor-quality data leads to poor decisions
- High implementation cost
- Security and privacy concerns
- Requires skilled users

Future of Decision Support Systems:

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Cloud Computing
- Big Data Analytics
- Internet of Things (IoT)
- Explainable AI (XAI)

CONCLUSION

- DSS transforms raw data into meaningful insights.
- It helps organizations make faster, smarter, and more reliable decisions.
- As AI and big data continue to evolve, DSS will become even more powerful and essential across industries.



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THANK YOU